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1 An energetic and physiological perspective on the $S \leftrightarrow R$ switch

The coexistence of host and virus populations following an epidemic collapse can be explain by a switch between Sensitive (S) and Resistant (R) cells. While the model with fixed traits captures well the dynamics, we lack a mechanistic understanding and a proper trait inference of the $S \leftrightarrow R$ switch. The switch might not be driven by fixed traits but rather by dynamic states. Here we explore the possibility that the prevalence of S and R cells depends on the balance between viral pressure and the energetic costs associated with resistance.

We first assume that resistance has a phenotypic nature. The genomic modifications (i.e. the hypervariability of the SOC) might be involved in the resistance but as a secondary effect and due to epigenetic for example. We propose that the transition between S and R phenotypes can be viewed as a shift in cellular energy allocation. Let E denote the total energy available to a cell. For sensitive cells, energy is primarily allocated to growth and basal maintenance:

$$E_S = G_S + M, \tag{1}$$

where G_S is the energy invested in biomass production and M represents maintenance requirements. In contrast, resistant cells allocate an additional fraction of their energy budget to defense mechanisms:

$$E_R = G_R + M + D, \tag{2}$$

where D represents the energetic cost of resistance. Consequently, resistant cells are expected to exhibit lower growth rates than sensitive cells in the absence of viral infection or without any compensatory physiological changes to compensate for the cost of resistance:

$$\mu_R = \mu_S(1 - c), \tag{3}$$

where μ_S and μ_R are the growth rates of sensitive and resistant cells, respectively, and c is the relative cost of resistance:

$$c = \frac{D}{E} \tag{4}$$

Under viral pressure, the dynamics of sensitive hosts (S), resistant hosts (R), and free viruses (V) can be described as the following ode system:

$$\frac{dS}{dt} = \mu_S S - \phi SV - \alpha(V)S + \gamma(V)R, \tag{5}$$

$$\frac{dR}{dt} = \mu_R R + \alpha(V)S - \gamma(V)R, \tag{6}$$

$$\frac{dV}{dt} = \beta\phi SV - \delta V, \tag{7}$$

where ϕ is the viral adsorption rate, β is the burst size, and δ is the viral decay rate. Note that for simplicity μ_i are linear but should be replace by non-linear growth.

The functions $\alpha(V)$ and $\gamma(V)$ describe transitions between the sensitive and resistant phenotypic states. Rather than specifying the two transition rates independently, we assume a constant total switching rate ϵ, such that

$$\epsilon = \alpha(V) + \gamma(V). \tag{8}$$

Viral abundance determines the direction of switching. The transition from sensitivity to resistance increases with viral abundance according to

$$\alpha(V) = \epsilon \frac{V}{K_V + V}, \tag{9}$$

while the reverse transition from resistance to sensitivity becomes more likely when viral pressure is low,

$$\gamma(V) = \epsilon \frac{K_V}{K_V + V}. \tag{10}$$

With this formulation, the total transition rate remains constant, while the relative probability of switching toward resistance or sensitivity depends on viral abundance.

From an energetic perspective, the fitness of each phenotype can be approximated as the per-capita growth F_i rate of S and R. The resistant phenotype is then favored whenever:

$$F_R > F_S. \tag{11}$$

Substituting the complete fitness expressions yields to a critical viral abundance V_c. This threshold defines the conditions under which resistance becomes advantageous. When $V < V_c$, the energetic cost of resistance outweighs its benefits and sensitive cells are expected to dominate. Conversely, when $V > V_c$, the

mortality imposed by viral infection exceeds the cost of resistance and resistant cells become selectively favored.
In this framework, temperature can influence the system through its effects on cellular growth, viral infection rates, and the energetic cost of resistance. The critical threshold therefore becomes temperature dependent. A "thermodynamic" interpretation of the $S \leftrightarrow R$ transition emerges: resistance is not solely determined by viral exposure, but by the balance between the energetic costs of defense and the mortality risk associated with infection. Changes in temperature may alter this balance by modifying energy allocation strategies, thereby reshaping the long-term coexistence between hosts and viruses.

2 Experimental tracking of the $S \leftrightarrow R$ switch using flow cytometry

If the transition between sensitive and resistant cells corresponds to a reversible physiological state rather than a stable genetic mutation, then the dynamics of the system might be observable at the single-cell level through flow cytometry. In this framework, resistance is interpreted as a shift in cellular resource allocation, whereby cells redirect energy from biomass production toward maintenance and defense processes. Consequently, resistant cells are expected to exhibit physiological signatures that differ from those of sensitive cells.

The proposed model predicts that the emergence of resistance should not only affect population abundances but also reshape the distribution of cellular phenotypes. During the course of an infection, a previously homogeneous population may become increasingly heterogeneous as cells adopt alternative energetic strategies. Flow cytometry therefore provides a powerful approach to quantify the appearance, persistence, and disappearance of these phenotypic states.
Several physiological markers can be used to characterize this transition.

2.1 Cellular morphology and photosynthetic state

Standard flow cytometric parameters, including forward scatter (FSC), side scatter (SSC), and chlorophyll autofluorescence, should be recorded throughout the experiment. These measurements provide information on cell size, internal complexity, and photosynthetic status, and may reveal the emergence of physiologically distinct subpopulations during infection and recovery.

2.2 Metabolic activity

Cellular metabolic activity can be quantified using fluorogenic esterase substrates such as CFDA or FDA. These probes provide an estimate of global cellular activity and may reveal differences in energy allocation between sensitive and resistant cells. According to the energetic framework developed above, resistant cells are expected to invest a larger fraction of their resources into maintenance and defense, potentially resulting in altered metabolic profiles compared with rapidly growing sensitive cells.

2.3 Energy storage

Neutral lipid content can be measured using BODIPY 505/515 or Nile Red staining. Changes in intracellular energy reserves may provide insight into how cells redistribute resources during the transition toward resistance. Resistant cells may either accumulate storage compounds as a protective strategy or consume reserves to sustain costly defense mechanisms.

2.4 Oxidative state

Reactive oxygen species (ROS) can be quantified using fluorescent probes such as H_2DCFDA or CellROX. Because defense activation and cellular maintenance are often associated with changes in redox homeostasis, resistant cells may exhibit distinct oxidative signatures. Monitoring ROS levels may therefore provide indirect evidence of the energetic costs associated with resistance.

2.5 Phenotypic structure and entropy

Beyond changes in average cellular properties, the proposed framework predicts that viral infection should alter the overall structure of the phenotypic landscape. Single-cell measurements can be projected into a multidimensional trait space using clustering or dimensionality reduction approaches such as UMAP, FlowSOM, or Phenograph. The appearance of a distinct cluster during epidemic progression would provide evidence for the emergence of a resistant physiological state.

Phenotypic heterogeneity can be quantified using Shannon entropy,

$$H = -\sum_i p_i \ln p_i, \tag{12}$$

where p_i represents the proportion of cells occupying phenotypic state i. Under this framework, entropy is expected to be low in an initially homogeneous sensitive population, increase during the transition phase as alternative states emerge, and stabilize once host-virus coexistence is established.

Together, these measurements provide a direct experimental approach to test the hypothesis that the $S \leftrightarrow R$ transition reflects a reversible shift in cellular energy allocation. By linking physiological markers, phenotypic heterogeneity, and population dynamics, flow cytometry offers a means to quantify the energetic basis of resistance and its role in shaping long-term host-virus coexistence.